# Assignment No. 4

## Title

**Survey Report on Types of Blockchains and Their Real-Time Use Cases**

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## 1. Aim

**To study the different types of Blockchain networks and understand their real-time applications and use cases in various industries.**

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## 2. Introduction

Blockchain is a distributed ledger technology that records transactions in a secure, transparent, and tamper-resistant manner. Instead of storing data at a single central location, blockchain maintains copies of the ledger across multiple computers called nodes.

Blockchain technology was initially introduced for cryptocurrency transactions, but its applications have expanded to areas such as banking, healthcare, supply chain management, education, government, insurance, and digital identity.

Blockchain networks can be classified according to who can access the network, who can participate in validating transactions, and how the network is governed.

The major types of blockchain are:

1. Public Blockchain
2. Private Blockchain
3. Consortium / Federated Blockchain
4. Hybrid Blockchain

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# 3. Types of Blockchain

## 3.1 Public Blockchain

A **Public Blockchain** is an open blockchain network in which anyone can participate. Users can generally read blockchain data, submit transactions, and, depending on the network, participate in transaction validation.

### Features

- Open and permissionless participation
- Decentralized network
- High transparency
- Transactions are recorded on a distributed ledger
- Uses consensus mechanisms for transaction validation
- Suitable for applications requiring public verification

### Examples

- Bitcoin
- Ethereum

### Real-Time Use Cases

**1. Cryptocurrency:**
Bitcoin uses blockchain technology to enable peer-to-peer digital currency transactions without requiring a central authority.

**2. Smart Contracts:**
Ethereum provides a platform for deploying smart contracts and decentralized applications (DApps).

**3. Decentralized Finance (DeFi):**
Blockchain-based applications can provide financial services such as lending, borrowing, and decentralized exchanges.

**4. Digital Assets:**
Public blockchains can be used to create and transfer digital assets and tokens.

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# 3.2 Private Blockchain

A **Private Blockchain** is a permissioned blockchain controlled by a particular organization or authorized group. Only approved participants can access the network or perform specific activities.

## Features

- Permission-based access
- Controlled by an organization
- Faster transaction processing can be achieved
- Greater privacy compared with public blockchains
- Participants are known and authorized

## Real-Time Use Cases

**1. Banking:**
Banks can use permissioned blockchain networks for internal transaction processing and record sharing.

**2. Supply Chain Management:**
Companies can use private blockchain networks to track products and transactions among authorized participants.

**3. Healthcare:**
Authorized healthcare organizations can share patient-related information securely while controlling access.

**4. Enterprise Record Management:**
Organizations can maintain shared records among departments or approved business partners.

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# 3.3 Consortium / Federated Blockchain

A **Consortium Blockchain** is a permissioned blockchain governed by multiple organizations rather than a single organization.

In this type of blockchain, a group of trusted organizations jointly manages the network.

## Features

- Controlled by multiple organizations
- Permissioned participation
- Suitable for business-to-business applications
- Provides controlled data sharing
- Governance is shared among participating organizations

### Real-Time Use Cases

**1. Banking and Financial Services:**
Multiple banks or financial institutions can share transaction or settlement information through a common permissioned network.

**2. Supply Chain:**
Manufacturers, suppliers, distributors, and retailers can share product information and transaction records.

**3. Insurance:**
Insurance companies and related organizations can share verified information for processing claims.

**4. Trade Finance:**
Multiple organizations involved in international trade can exchange and verify transaction documents.

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## 3.4 Hybrid Blockchain

A **Hybrid Blockchain** combines features of public and private blockchains. Some information or operations can be controlled privately, while selected information can be made publicly verifiable.

### Features

- Combination of public and private blockchain features
- Controlled access to sensitive information
- Selected information can be publicly verified
- Provides flexibility for organizations
- Suitable where both privacy and transparency are required

### Real-Time Use Cases

**1. Healthcare:**
Sensitive patient information can remain restricted while selected information or verification records can be shared with authorized parties.

**2. Government Services:**
Government organizations can keep confidential records private while making selected records publicly verifiable.

**3. Supply Chain:**
Business-sensitive information can remain private while product authenticity or selected tracking information can be verified by customers.

# 4. Comparison of Different Types of Blockchain

| Parameter | Public | Private | Consortium | Hybrid |
|---|---|---|---|---|
| Access | Open | Restricted | Restricted | Partly open / restricted |
| Control | Decentralized | Single organization | Multiple organizations | Combination |
| Participants | Anyone | Authorized users | Selected organizations | Depends on application |
| Transparency | High | Controlled | Controlled | Selective |
| Privacy | Relatively low | High | High | High |
| Governance | Community/network | Single organization | Multiple organizations | Combination |
| Transaction Speed | Generally lower | Generally higher | Generally higher | Depends on design |
| Main Applications | Cryptocurrency, DApps | Enterprise systems | Inter-organizational systems | Enterprise + public verification |

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# 5. Real-Time Blockchain Use Cases

## 5.1 Banking and Finance

Blockchain can be used for recording and transferring financial assets, facilitating settlement, and developing decentralized financial applications.

**Examples:**

- Digital payments
- Cross-border transactions
- Asset settlement
- DeFi applications
- Digital currencies

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## 5.2 Supply Chain Management

Blockchain can maintain a shared record of products as they move through different stages of the supply chain.

**Example:**

**Manufacturer → Distributor → Warehouse → Retailer → Customer**

Blockchain can help participants verify information such as product origin, movement, and transaction history.

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## 5.3 Healthcare

Blockchain can be used to maintain controlled and auditable records related to healthcare information.

**Applications:**

- Medical record sharing
- Drug traceability
- Healthcare data management
- Verification of credentials

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## 5.4 Education

Blockchain can be used to create verifiable digital records of academic achievements.

**Applications:**

- Digital certificates
- Degree verification
- Student academic records
- Credential verification

A university or educational institution can issue a digital certificate that can later be verified without relying entirely on manual document verification.

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## 5.5 Government

Blockchain can support applications where multiple parties need to verify records.

**Applications:**

- Digital identity
- Record management
- Document verification
- Public-service applications
- Asset or property records

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## 5.6 Real Estate

Blockchain can maintain records related to property ownership and transactions.

**Applications:**

- Property records
- Ownership verification
- Digital agreements
- Transaction history

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## 5.7 Intellectual Property and Digital Content

Blockchain can record ownership or transaction information related to digital assets.

**Applications:**

- Digital ownership records
- Copyright management
- Digital collectibles
- Royalty tracking

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# 6. Advantages of Blockchain

1. **Decentralization** – Data is distributed among multiple participants.
2. **Transparency** – Transactions can be independently verified according to the network's access rules.
3. **Security** – Cryptographic techniques help protect blockchain records.
4. **Immutability** – Confirmed records are designed to be difficult to alter.

5. **Traceability** – Transactions and asset movements can be tracked.
6. **Automation** – Smart contracts can automatically execute predefined rules.
7. **Reduced Intermediaries** – Some applications can reduce the need for intermediaries.

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# 7. Limitations of Blockchain

1. Some blockchain networks can have scalability limitations.
2. Transaction fees can vary depending on network conditions.
3. Public blockchains may require significant computational resources depending on their consensus mechanism.
4. Privacy can be a concern when information is publicly visible.
5. Implementation and maintenance can be technically complex.

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# 8. Survey Findings

From the survey, it can be observed that different blockchain types are designed for different requirements.

- **Public Blockchain** is suitable when open participation and public verification are important.
- **Private Blockchain** is suitable when an organization requires controlled access and privacy.
- **Consortium Blockchain** is suitable when multiple organizations need to jointly manage and share records.
- **Hybrid Blockchain** is suitable when an application requires both controlled information and selected public verification.

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# 9. Conclusion

Blockchain technology has applications beyond cryptocurrency. Different types of blockchain provide different levels of access, control, privacy, and transparency.

Public, private, consortium, and hybrid blockchains can be used for different real-world requirements. Applications can be found in banking, finance, supply chain management, healthcare, education, government, real estate, and digital asset management.

Thus, blockchain is a useful technology for applications where **shared records, verification, traceability, security, and controlled data sharing** are important.

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